02 (a-i) TECHNIQUES IN AVOIDING AMBIGUITY: TEMPLATE-BASED

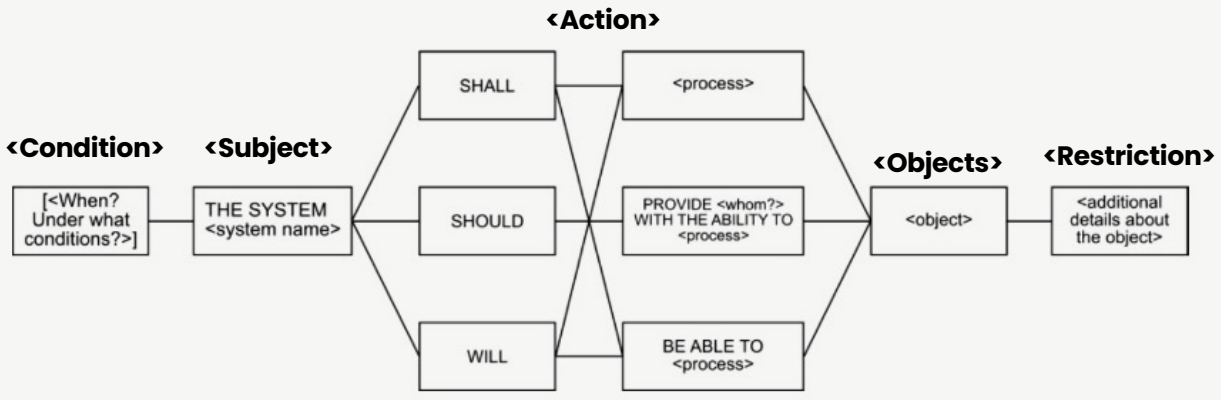
- *PHRASE TEMPLATES*

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Requirements Template

TEMPLATE-BASED: PHRASE TEMPLATES



innovative ● entrepreneurial ● global

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Steps to write single requirements using template

TEMPLATE-BASED: PHRASE TEMPLATES

01 Determine legal obligation (shall, should or will)

02 Determine the core of the requirement (process verb)

03 Characterize the activity of the system

04 Insert objects

05 Determine logical and temporal conditions

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Steps to write single requirements using template

TEMPLATE-BASED: PHRASE TEMPLATES

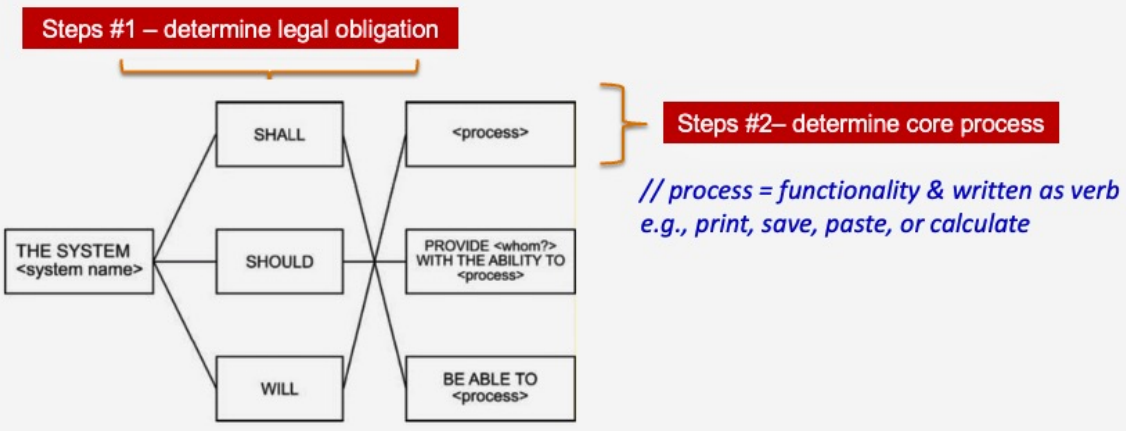


Figure 5-2 *The core of a requirement and its legal obligation* (eBook, pg. 55)

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Steps to write single requirements using template: Step #1

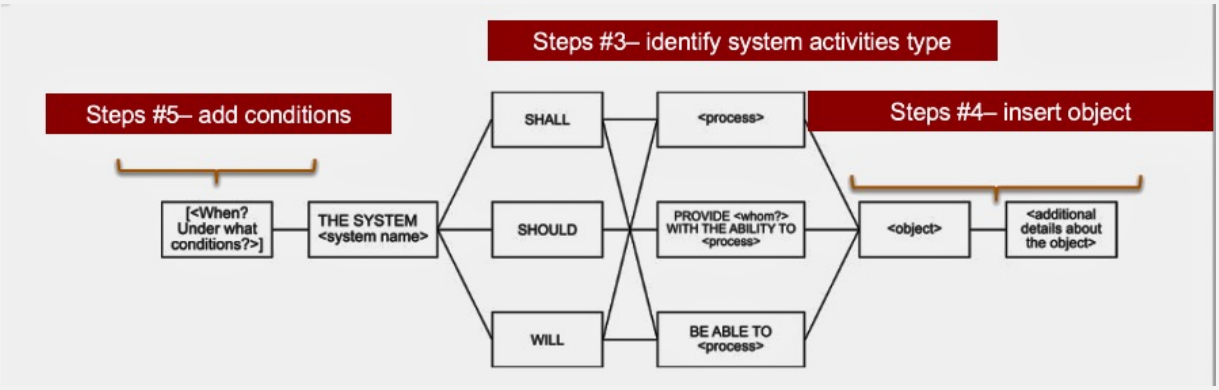
TEMPLATE-BASED: PHRASE TEMPLATES

Verb	(Legal) Obligation	Explanation
shall	Duty / Required	Mandatory, Legally binding, "must have"
should	Wish	Strongly advised but not mandatory
will (or verb in present tense without auxiliary verb)	Intention	Factual statement not considered a requirement
may	Recommendation	Desirable but not necessary

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Steps to write single requirements using template

TEMPLATE-BASED: PHRASE TEMPLATES



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Steps to write single requirements using template: Steps #3 - #5

TEMPLATE-BASED: PHRASE TEMPLATES

Type of System Activities	Explanation	Phrase Templates
Ubiquitous Requirements	Must always hold	The <system name> shall <system response>
Event-driven Requirements	Triggered by an external event	WHEN <optional preconditions> <trigger> the <system name> shall <system response>
Unwanted Behaviour	Describing situations to be avoided	IF <optional preconditions> <trigger>, THEN the <system name> shall <system response>
State-driven Requirements	Apply only in certain states	WHILE <in a specific state> the <system name> shall <system response>
Optional Features	Only if some feature is included in the system	WHERE <feature is included> the <system name> shall <system response>

Steps to write single requirements using template: Steps #3 - #5

TEMPLATE-BASED: PHRASE TEMPLATES

Type of System Activities	Phrase Templates	Example
Ubiquitous Requirements	The <system name> shall <system response>	The library platform shall prevent unauthorized access to member data
Event-driven Requirements	WHEN <optional preconditions> <trigger> the <system name> shall <system response>	WHEN the date is the 31st. December of each year, between 8:00pm and 11:59pm GMT+8, the library platform shall archive all inactive lending data older than 365 days
Unwanted Behaviour	IF <optional preconditions> <trigger>, THEN the <system name> shall <system response>	IF an item is reserved and the lending period of the item is expired, then the library platform shall notify the member who borrowed the item via email
State-driven Requirements	WHILE <in a specific state> the <system name> shall <system response>	WHILE an item is lent, the library platform shall prevent members from requesting it
Optional Features	WHERE <feature is included> the <system name> shall <system response>	WHERE eBook feature is included, the library platform shall give members access to borrow eBooks.

02 (b) TECHNIQUES IN AVOIDING AMBIGUITY: GLOSSARY

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Why use a Glossary

- Glossaries are a core means of establishing shared understanding of the terminology used.
- They help avoid people involved as stakeholders or developers using and interpreting the same terms in different ways.

GLOSSARY

Rules

1. Manage the glossary centrally
2. Maintain the glossary over the entire course of the system development
3. Define a person or small group who is responsible for the glossary
4. Use a uniform style and structure for the glossary
5. Involve the stakeholders and seek agreement about the terminology
6. Make the glossary accessible for everybody involved.
7. Make the use of the glossary mandatory.
8. Check work products for proper glossary usage.

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Glossary Content

GLOSSARY

01 Definitions for all terms relevant for the system
- Context-specific technical terms
- Everyday concepts that have a special meaning in the given context

02 Abbreviations and acronyms

Content

03 Synonyms
- Different terms with same meaning

04 Homonyms
- Identical terms with different meanings (should be avoided if possible)

Example: IREB CPRE Glossary

GLOSSARY

Term is defined in same glossary

Redundancy*	occurrence of the same information or resource.
Release*	uration that has been released for installation and use by customers.
Reliability*	The degree to which a System performs specified functions under specified conditions for a specified period of time. **Note:** Reliability may be stated as a Quaality requirement.
Requirements analysis	1. A need perceived by a Stakeholder. 2. A capability or property that a System shall have. 3. A documented representation of a need, capability orpoperty.
Requirements baseline	1. Analysis of elicited Requirements Engineering.
Requirements branching	A baseline for a set of Requirements.
Requirements configuration	→ Branch
Requirements conflict*	→ Configuration
Requirements conflict*	1. A situation where two or more Requirements cannot be satisfied together. 2. A situation where two or more Stakeholders disagree about certain

Synonym

THANK YOU

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